

Homework problemsSOLUTIONS

① (a) $\frac{dx}{dt} = -\lambda x$, $\lambda = \text{const}$, $x(0) = x_0$.

$$\frac{dx}{x} = -\lambda dt$$

$$\int \frac{dx}{x} = -\lambda \int dt$$

$$\ln|x| = -\lambda t + C$$

$$|x| = e^C e^{-\lambda t}$$

$$x = \underbrace{\pm e^C}_{\text{rename } C} e^{-\lambda t}$$

$$x = C e^{-\lambda t}$$

Using $x(0) = x_0$:

$$x_0 = C e^0 \Rightarrow C = x_0$$

$$\Rightarrow \underline{x(t) = x_0 e^{-\lambda t}}$$

(b) $y' + y \tan x = \cos x$

This is a 1st-order linear differential equation. It can be solved by either of the two methods: method of a varying constant, or the integrating factor method. Let's use the former.

Step 1: solve the homogeneous equation

$$\frac{dy}{dx} + y \tan x = 0$$

$$\frac{dy}{dx} = -y \tan x$$

$$\frac{dy}{y} = -\tan x \, dx$$

$$\int \frac{dy}{y} = - \int \frac{\sin x \, dx}{\cos x}$$

$$-\sin x \, dx = d(\cos x)$$

$$\ln y = \int \frac{d(\cos x)}{\cos x}$$

$$\ln y = \ln(\cos x) + C$$

$$y = \underbrace{e^C}_{\text{rename } C} e^{\ln(\cos x)}$$

$$y = C \cos x$$

Step 2 : replace constant C with function $C(x)$

$$y = C(x) \cos x$$

Substitute into the original equation:

$$C'(x) \cos x - \cancel{C(x) \sin x} + \cancel{C(x) \cos x} \frac{\sin x}{\cos x} = \cos x$$

$$\frac{dC}{dx} \cos x = \cos x$$

$$\frac{dC}{dx} = 1$$

$$\int dC = \int dx$$

$$C(x) = x + C_1$$

Therefore

$$y = (x + C_1) \cos x$$

$$y = \underbrace{x \cos x}_{\text{particular integral}} + \underbrace{C_1 \cos x}_{\text{complementary function}}$$

particular integral complementary function

(c) $y'' + 4y' + 5y = \sin 2x$

(3)

This is a 2nd-order linear DE. Solving the homogeneous equation first:

$$y'' + 4y' + 5y = 0 \quad (*)$$

Seek solution in the form

$$y = e^{sx}$$

$$y' = se^{sx}, \quad y'' = s^2 e^{sx}$$

Substituting back into (*):

$$s^2 e^{sx} + 4se^{sx} + 5e^{sx} = 0 \quad \left. \vphantom{s^2 e^{sx} + 4se^{sx} + 5e^{sx} = 0} \right\} \times e^{-sx}$$

$$s^2 + 4s + 5 = 0 \quad - \text{auxiliary equation}$$

Roots:

$$s = \frac{-4 \pm \sqrt{16 - 20}}{2} = \frac{-4 \pm \sqrt{-4}}{2} = \frac{-4 \pm 2i}{2}$$

$$\Rightarrow s = -2 \pm i$$

The general solution of the homogeneous equation is

$$y = C_1 e^{(-2+i)x} + C_2 e^{(-2-i)x}$$

which can also be written in the 'real form'

$$y = e^{-2x} (C_1 \cos x + C_2 \sin x)$$

(with a different choice of C_1, C_2).

To solve the inhomogeneous equation, note that the function on the R.H.S. has the form

$$e^{ax} [P_n(x) \cos bx + Q_m(x) \sin bx]$$

with $a = 0$, $P_n(x) = 0$, $Q_m(x) = 1$, $b = 2$.

Hence, we can seek particular integral in

the form: $y_p = A \cos 2x + B \sin 2x$

[$a \pm ib = \pm 2i$ is not among the roots of the auxiliary equation; A and B are polynomials of degree zero (as $Q_m(x)$), with undetermined coefficients, i.e. A and B .] (4)

$$y_p' = -2A \sin 2x + 2B \cos 2x$$

$$y_p'' = -4A \cos 2x - 4B \sin 2x$$

Substituting back into the differential equation:

$$\begin{aligned} & \underline{-4A \cos 2x} - \underline{4B \sin 2x} + 4(\underline{-2A \sin 2x} + \underline{2B \cos 2x}) \\ & + 5(\underline{A \cos 2x} + \underline{B \sin 2x}) = \sin 2x \end{aligned}$$

$$(-4A + 8B + 5A) \cos 2x + (-4B - 8A + 5B) \sin 2x = \sin 2x$$

$$(A + 8B) \cos 2x + (B - 8A) \sin 2x = \sin 2x$$

$$\begin{cases} A + 8B = 0 \\ B - 8A = 1 \end{cases} \Rightarrow \begin{aligned} & A = -8B \\ & \text{substitute into 2nd eq.} \end{aligned}$$

$$B + 64B = 1 \quad \Leftrightarrow \quad 65B = 1$$

$$\Rightarrow B = \frac{1}{65}, \quad A = -\frac{8}{65}$$

$$\Rightarrow y_p = -\frac{8}{65} \cos 2x + \frac{1}{65} \sin 2x$$

So, the general solution of the inhomogeneous equation is:

$$y = \underbrace{C_1 e^{-2x} \cos x + C_2 e^{-2x} \sin x}_{\text{complementary function}} - \frac{8}{65} \cos 2x + \frac{1}{65} \sin 2x$$

particular
integral

(2) $\frac{d^2 y}{dx^2} + \lambda y = 0$, $y(0) = y(1) = 0$. (5)

This is a linear 2nd-order differential equation with constant coefficients. We seek its solution in the form:

$$y = e^{sx}$$

$$y' = se^{sx} , \quad y'' = s^2 e^{sx} \quad \text{-- substituting:}$$

$$s^2 e^{sx} + \lambda e^{sx} = 0 \quad \left\} \times e^{-sx}$$

$$s^2 + \lambda = 0$$

$$s^2 = -\lambda$$

$$s = \pm \sqrt{-\lambda}$$

General solution:

$$y = C_1 e^{\sqrt{-\lambda} x} + C_2 e^{-\sqrt{-\lambda} x}$$

$\left\} \begin{array}{l} \text{we assume} \\ \lambda \neq 0 \end{array} \right.$

Implementing the boundary condition:

$$y(0) = 0 \quad \Rightarrow \quad C_1 + C_2 = 0 \quad \Rightarrow \quad C_2 = -C_1 \equiv -C$$

$$y = C (e^{\sqrt{-\lambda} x} - e^{-\sqrt{-\lambda} x})$$

2nd boundary condition $y(1) = 0$

$$C (e^{\sqrt{-\lambda}} - e^{-\sqrt{-\lambda}}) = 0 \quad (*)$$

For $\lambda < 0$ the exponents are real, and $e^{\sqrt{-\lambda}} > 1$, while $e^{-\sqrt{-\lambda}} < 1$. So (*) can only be satisfied for $C = 0$ and we only have a trivial solution $y = 0$.

For $\lambda > 0$

$$\sqrt{-\lambda} = i\sqrt{\lambda}$$

(6)

So

$$y = C (e^{i\sqrt{\lambda}x} - e^{-i\sqrt{\lambda}x})$$

which can be written as

$$y = C \sin(\sqrt{\lambda}x)$$

(with a different C).

Recall:

$$e^{i\theta} = \cos\theta + i\sin\theta$$

$$e^{-i\theta} = \cos\theta - i\sin\theta$$

$$\cos\theta = \frac{e^{i\theta} + e^{-i\theta}}{2}$$

$$\sin\theta = \frac{e^{i\theta} - e^{-i\theta}}{2i}$$

Now, $y(1) = 0$ gives

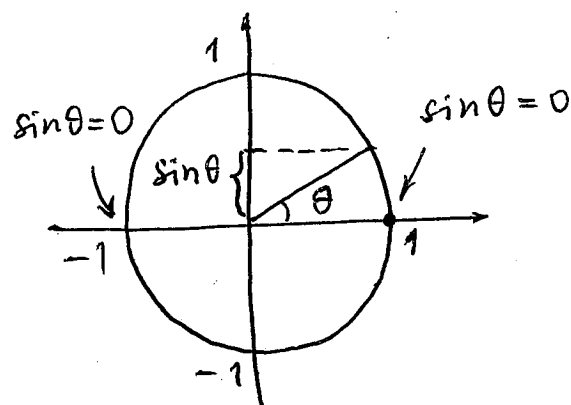
$$C \sin(\sqrt{\lambda}) = 0$$

So:

$$\sqrt{\lambda} = \pi n$$

$$(n = 1, 2, \dots)$$

$$\boxed{\lambda = \pi^2 n^2}$$



$\sin\theta = 0$ for

$$\theta = \pi n, n \in \mathbb{Z}$$

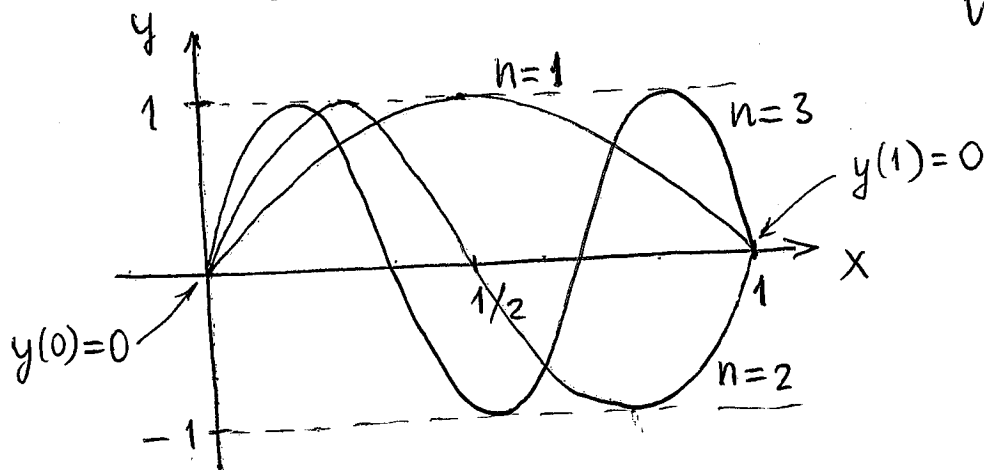
For these values of λ
the differential equation

$y'' + \lambda y = 0$ with boundary conditions $y(0) = y(1) = 0$
has nontrivial solutions:

$$y = C \sin(\pi n x)$$

Sketching the first few (for $0 \leq x \leq 1$)

We use $C = 1$.



- ③ (a) $u_x u_y - 3u = x$ 1st order, nonlinear
 (b) $y u_{xxy} - e^x u_x + 3 = 0$ 3rd order, linear inhomogeneous
 (c) $\rho u_{tt} + \mu u_{xxxx} = 0$ 4th order, linear homogeneous
 (d) $u u_{xy} - u_x u_y = 0$ 2nd order, nonlinear
 (e) $u_{xx} + u_{yy} + \log u = \log x$ 2nd order, nonlinear (because of $\log u$ term)

④ $u = f(x^2 + y^2)$

$$u_x = f'(x^2 + y^2) \frac{\partial(x^2 + y^2)}{\partial x} \quad \left. \vphantom{\frac{\partial(x^2 + y^2)}{\partial x}} \right\} \text{using chain rule}$$

(1) $u_x = f'(x^2 + y^2) 2x$

$$u_y = f'(x^2 + y^2) \frac{\partial(x^2 + y^2)}{\partial y}$$

(2) $u_y = f'(x^2 + y^2) 2y$

Dividing (1) by (2) to eliminate f' , we obtain

$$\frac{u_x}{u_y} = \frac{2x}{2y}$$

$$y u_x = x u_y$$

$$\underline{y u_x - x u_y = 0}, \text{ as required.}$$

⑤ (a) $u = f(xy)$

(1) $u_x = f'(xy) \frac{\partial}{\partial x}(xy) = f'(xy) y$

(2) $u_y = f'(xy) \frac{\partial}{\partial y}(xy) = f'(xy) x$

Dividing (1) by (2) to eliminate $f'(xy)$:

$$\frac{u_x}{u_y} = \frac{y}{x} \quad \Leftrightarrow \quad \underline{x u_x - y u_y = 0.}$$

$$(b) \quad u = f(x^2 + y^2) + g(x)$$

$$u_x = f'(x^2 + y^2) \frac{\partial(x^2 + y^2)}{\partial x} + g'(x)$$

$$u_x = f'(x^2 + y^2) 2x + g'(x) \quad (1)$$

$$u_y = f'(x^2 + y^2) 2y \Rightarrow f'(x^2 + y^2) = \frac{u_y}{2y}$$

Substituting this into (1) to eliminate $f'(x^2 + y^2)$:

$$u_x = \frac{x u_y}{y} + g'(x) \quad (2)$$

To get rid of $g'(x)$, differentiate (2) with respect to y :

$$u_{xy} = x \frac{u_{yy}}{y} - x \frac{u_y}{y^2} \quad \} \times y^2$$

$$y^2 u_{xy} = xy u_{yy} - x u_y,$$

or

$$\underline{y^2 u_{xy} - xy u_{yy} + x u_y = 0.}$$

(6) (a) $u_{xy} = 1$

Integrate with respect to y :

$$\int u_{xy} dy = \int dy$$

$$u_{xx} = y + f(x)$$

↑
arbitrary function ("constant" with respect to y).

Integrating over x :

$$\int u_{xx} dx = \int y dx + \int f(x) dx$$

$$u_x = xy + f_1(x) + g(y)$$

↑
integral of an arbitrary function of x is another arbitrary function of x

← "constant" with respect to x

Integrating over dx again:

(9)

$$\int u_x dx = \int xy dx + \underbrace{\int f_1(x) dx}_{\text{arb. function } h(x)} + \int g(y) dx$$

$$u = \frac{1}{2} x^2 y + x g(y) + h(x) + \underbrace{f(y)}_{\text{again, an arbitrary "constant" with respect to } x}$$

again, an arbitrary "constant" with respect to x .

So: $u = \frac{1}{2} x^2 y + x g(y) + f(y) + h(x),$

where f, g and h are arbitrary functions.

(8) $u_{xx} - u = 0$ (*)

A similar 2nd-order ordinary differential equation $u'' - u = 0$ for $u(x)$ has the

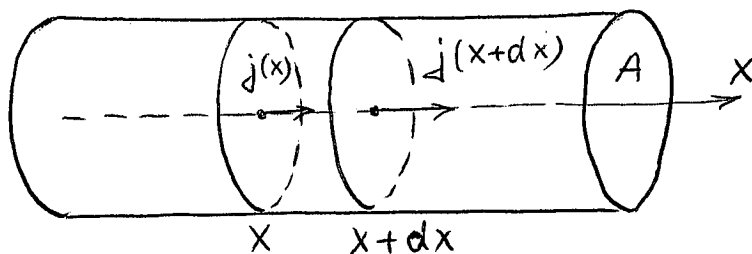
general solution $u = C_1 e^x + C_2 e^{-x}$.

Hence, (*) has the solution

$$u = \frac{f(y) e^x + g(y) e^{-x}}{1},$$

where $f(y)$ and $g(y)$ are arbitrary functions of y (i.e. "constants" with respect to x).

(7)



Let q be the heat energy density, i.e. heat energy per unit volume.

The amount of heat between the two planes at x and $x+dx$ is

$$q \underbrace{A dx}_{\text{volume of the "slice"}}$$

The amount of heat that flows into the slice in time dt is $j(x) A dt$, where $j(x)$ is the heat energy flux density at x , and A is the cross section area of the rod. (10)

The amount of heat that flows out of the slice is $j(x+dx) A dt$ (in time dt).

Since the heat energy is conserved, the change of the heat energy in the slice over the time interval dt can be related to $j(x)$ and $j(x+dx)$, as:

$$\underbrace{\frac{\partial}{\partial t} (q A dx)}_{\text{change in the amount of heat energy in the slice}} dt = j(x) A dt - j(x+dx) A dt$$

$$\frac{\partial q}{\partial t} A dx = - [j(x+dx) - j(x)] A$$

$$\frac{\partial q}{\partial t} = - \frac{\partial j}{\partial x}$$

$$\frac{\partial q}{\partial t} + \frac{\partial j}{\partial x} = 0 \quad (1) \quad \text{energy conservation.}$$

The heat energy density q is proportional to the temperature at x : $q = C \rho T$, where ρ is the density of the rod, and C is the specific heat. The flux density is proportional to the temperature gradient (Fourier's law):

$$j = -\alpha \frac{\partial T}{\partial x} \quad (2)$$

where α is the thermal conductivity. (Minus in this equation is a consequence of heat flowing from planes with higher T to those with lower T .)

Substituting (2) and the expression for g into (11) Eq. (1), we have:

$$\frac{\partial}{\partial t}(C\rho T) - \frac{\partial}{\partial x}\left(x \frac{\partial T}{\partial x}\right) = 0$$

For a uniform rod $C = \text{const}$, $\rho = \text{const}$, $x = \text{const}$,

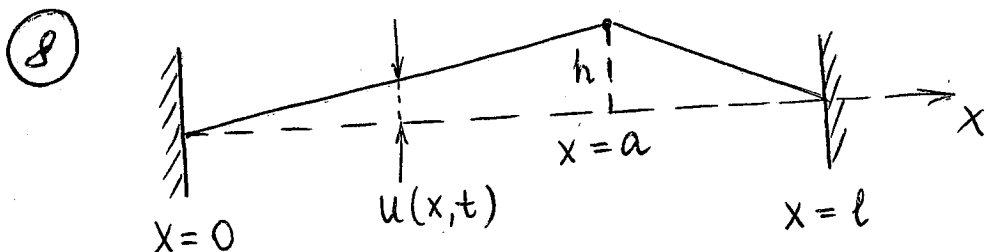
hence:

$$C\rho \frac{\partial T}{\partial t} - x \frac{\partial^2 T}{\partial x^2} = 0,$$

or

$$\boxed{\frac{\partial u}{\partial t} - K \frac{\partial^2 u}{\partial x^2} = 0}$$

where $K = \frac{x}{C\rho}$, and we denoted the temperature u .



The boundary conditions for the displacement of the string $u(x,t)$ are: $u(0,t) = 0$ (fixed ends), $u(l,t) = 0$.

At $t=0$ the string has the shape shown on the diagram. This can be written analytically as two linear functions:

$$u(x,0) = \begin{cases} C_1 x, & 0 \leq x \leq a \\ C_2 (l-x), & a \leq x \leq l \end{cases}$$

C_1 and C_2 are chosen so that $u(a,0) = h$, so that:

$$u(x,0) = \begin{cases} h \frac{x}{a}, & 0 \leq x \leq a \\ h \frac{l-x}{l-a}, & a \leq x \leq l. \end{cases}$$

At $t=0$ the string is at rest, i.e. has zero velocity:

$$u_t(x,0) = 0.$$

This and $u(x,0)$ above are the initial conditions.